

A CLOSED FORM AND A LUCAS CONGRUENCE FOR APÉRY'S COMPANION SEQUENCE

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ABSTRACT. We give a two-term closed form for the numerators b_n of Apéry's approximations to $\zeta(3)$, a Lucas-type congruence $p^3 b_{ap+r} \equiv b_a a_r \pmod{p}$ for them, and a correction to a published integrality claim. Every statement is proved in Lean 4 and checked by its kernel; each is followed by the name of the declaration that proves it.

1. INTRODUCTION

Apéry's proof that $\zeta(3) \notin \mathbb{Q}$ rests on the pair of sequences $a_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$ and a companion b_n with $b_n/a_n \rightarrow \zeta(3)$. The first sequence is arithmetically well understood: it satisfies a Lucas congruence $a_{ap+r} \equiv a_a a_r \pmod{p}$ (Gessel [2]), so it respects base- p digits. The companion b_n has attracted no such statement, and it is usually written with a nested alternating inner sum, which makes it awkward to handle modulo p .

We report three things.

First, a closed form (Theorem 3.4): b_n is the same binomial sum as a_n weighted by a *two-term* harmonic expression,

$$b_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 (2H_n^{(3)} - H_k^{(3)}), \quad H_m^{(r)} = \sum_{j=1}^m j^{-r}.$$

The nested sum disappears.

Second, a Lucas-type congruence for the companion itself (Theorem 3.5):

$$p^3 b_{ap+r} \equiv b_a a_r \pmod{p} \quad (a, r < p).$$

The factor p^3 is the weight of $\zeta(3)$. The same argument gives the analogous congruence for Franel's companion at weight 2 (Theorem 3.6), and for the first solution of the Brown–Zudilin family (Theorem 2.4).

Third, a correction. Brown and Zudilin [1] record $d_n^5 P_n \in \mathbb{Z}$ for the numerators of their cellular approximations to $\zeta(5)$, where $d_n = \text{lcm}(1, \dots, n)$. Their own printed value $P_2 = 1190161/384$ does not satisfy this: the defect is a factor $12 = 2^2 \cdot 3$, and 12 is minimal (Theorem 4.2).

A second, benign slip in the same author's earlier work is worth recording, though we do not formalise it: equation (8.7) of [5], defining the rational function $R(t)$, omits the factor $(h_0 + 2t)$ present in (8.2). That factor is the only one antisymmetric under $t \mapsto -t - h_0$, and it is what forces the coefficients of the even zeta values to vanish; without it the parity structure of the linear forms is reversed.

Throughout, p is prime, and for $x, y \in \mathbb{Q}$ we write $x \equiv y \pmod{p}$ to mean $v_p(x-y) \geq 1$.
PInt, PDvd, PCong

2. FIRST SOLUTIONS

Definition 2.1.

$$A(n, k) = \binom{n}{k}^2 \binom{n+k}{k}^2, \quad a_n = \sum_{k=0}^n A(n, k),$$

$$T(n, k, l) = \binom{n+k}{n} \binom{n}{k}^2 \binom{n+l}{n} \binom{n}{l}^2 \binom{n+k+l}{n}, \quad Q_n = \sum_{k=0}^n \sum_{l=0}^n T(n, k, l).$$

A, apery, T, Q

Lemma 2.2. For $r, s < p$ one has $\binom{ap+r}{bp+s} \equiv \binom{a}{b} \binom{r}{s} \pmod{p}$; and $\binom{r+s}{s} \equiv 0 \pmod{p}$ when $s \leq r$ and $r+s \geq p$. choose_digits, choose_carry_zero

Lemma 2.3. For $r, s < p$, $A(ap+r, bp+s) \equiv \binom{a}{b}^2 \binom{a+b}{b}^2 A(r, s) \pmod{p}$; and for $r, s, t < p$, $T(ap+r, bp+s, cp+t) \equiv T(a, b, c) T(r, s, t) \pmod{p}$. A_digits, T_digits

Summing Lemma 2.3 over a full base- p block, the double sum factors:

Theorem 2.4. For $r < p$ and every $a \geq 0$,

$$a_{ap+r} \equiv a_a a_r \pmod{p}, \quad Q_{ap+r} \equiv Q_a Q_r \pmod{p},$$

and hence $a_n \equiv \prod_i a_{n_i}$, $Q_n \equiv \prod_i Q_{n_i} \pmod{p}$ for the base- p expansion $n = \sum_i n_i p^i$. apery_lucas, Q_lucas, apery_lucas_digits, Q_lucas_digits

The statement for a_n is Gessel's [2]; the proof formalised here is the digit-block argument, and it applies verbatim to Q_n because the extra binomial factors in T satisfy the same carry dichotomy.

3. APÉRY'S COMPANION

Definition 3.1. $P(n) = 34n^3 + 51n^2 + 27n + 5$, and b_n is defined by

$$b_0 = 0, \quad b_1 = 6, \quad (n+2)^3 b_{n+2} = P(n+1) b_{n+1} - (n+1)^3 b_n.$$

Ppol, bApery

This is Apéry's recurrence; $b_2 = 351/4$ and $b_3 = 62531/36$.

Definition 3.2. $\tilde{b}_n = \sum_{k=0}^n A(n, k) (2H_n^{(3)} - H_k^{(3)})$.

Harm, bMin

Lemma 3.3. $(m+2)^3 \tilde{b}_{m+2} - P(m+1) \tilde{b}_{m+1} + (m+1)^3 \tilde{b}_m = 0$ for all $m \geq 0$. bMin_rec

The proof is a single telescoping step: applying the recurrence operator to the summand of $\tilde{b}$ produces an exact difference $\Psi(m, k+1) - \Psi(m, k)$, whose sum collapses to the boundary values $\Psi(m, 0) = \Psi(m, m+3) = 0$. The two Zeilberger-type certificates involved are polynomial identities in $\mathbb{Q}[m, k]$. star, star_pre

Theorem 3.4. $\tilde{b}_n = b_n$ for all n ; that is,

$$\sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 (2H_n^{(3)} - H_k^{(3)}) = b_n.$$

bMin_eq_bApery, apery_b_harmonic_closed_form

Theorem 3.5. For $a, r < p$, $p^3 b_{ap+r} \equiv b_a a_r \pmod{p}$.

bApery_lucas

The weight 3 enters as follows: modulo p the summands with $k \not\equiv 0$ contribute nothing after multiplication by p^3 , while the layer $k = jp$ reproduces $H^{(3)}$ at $\lfloor \cdot/p \rfloor$, since $p^3 \sum_{m \leq y, p|m} m^{-3} = \sum_{j \leq y/p} j^{-3}$. The remaining digit factorisation is Lemma 2.3.

Theorem 3.6. Let $f_n = \sum_k \binom{n}{k}^3$ and $\beta_n = \sum_k \binom{n}{k}^3 (\frac{1}{4} H_k^{(2)} + \frac{3}{4} (H_k^{(1)})^2 - \frac{3}{4} H_k^{(1)} H_{n-k}^{(1)})$. Then for $p \neq 2$ and $a, r < p$,

$$p^2 \beta_{ap+r} \equiv \beta_a f_r \pmod{p}.$$

franel, bFranel, bFranel_lucas

4. A DENOMINATOR DEFECT

Definition 4.1. $d_n = \text{lcm}(1, \dots, n)$, and $P_2 = 1190161/384$ is the value printed in [1].
`dlcm, BZ_P2`

Theorem 4.2. $d_2^5 P_2 = 1190161/12 \notin \mathbb{Z}$, while $12 d_2^5 P_2 = 1190161 \in \mathbb{Z}$; and if $c \in \mathbb{N}$ satisfies $c d_2^5 P_2 \in \mathbb{Z}$, then $12 \mid c$. *BZ_P2_not_integral, BZ_P2_factor_twelve, BZ_twelve_minimal*

Since $d_2 = 2$ and $1190161 \equiv 1 \pmod{12}$, the defect is exactly $2^2 \cdot 3$: the claim of [1] holds after multiplication by 12 and by no smaller factor.

5. THE DEVELOPMENT

Lean 4 (v4.33.0-rc1) with Mathlib cd580e54; ten modules, 2107 lines; no `sorry` and no `native.decide`. Every declaration cited above satisfies

`#print axioms` $\longrightarrow$ `[propext, Classical.choice, Quot.sound]`.

Build with `lake exe cache get && lake build`.

REFERENCES

- [1] F. Brown and W. Zudilin, *On cellular rational approximations to $\zeta(5)$* , preprint [arXiv:2210.03391v3](#) (2022).
- [2] I. M. Gessel, *Some congruences for Apéry numbers*, J. Number Theory **14** (1982), no. 3, 362–368.
- [3] The mathlib Community, *The Lean mathematical library*, CPP 2020, 367–381. [arXiv:1910.09336](#).
- [4] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, CADE 28, LNCS 12699, 625–635.
- [5] W. Zudilin, *Arithmetic of linear forms involving odd zeta values*, J. Théor. Nombres Bordeaux **16** (2004), 251–291.